

BREMERHAVEN, Germany

WMO: 101290

Lat: 53.5333N

Lon: 8.5761E

Elev: 5

StdP: 101.26

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-9.4	-7.0	-11.7	1.4	-7.8	-10.0	1.6	-6.2	17.9	8.1	15.3	6.3	3.7	80	0.674

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.3	27.2	18.9	25.1	18.0	23.1	17.4	20.1	25.0	19.2	23.2	18.3	22.0	4.4	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	13.4	22.0	17.6	12.6	21.2	16.7	11.9	20.3	57.7	25.1	54.5	23.2	51.6	22.1	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 13.7	12.2	10.9	DB	-9.1	31.1	4.1	2.2	-12.1	32.7	-14.5	33.9	-16.8	35.2	-19.8	36.7	(4)
(5)			WB	-9.5	22.3	4.0	0.9	-12.4	23.0	-14.8	23.5	-17.1	24.0	-20.0	24.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.5	1.9	1.7	4.9	7.9	12.6	15.3	17.5	17.4	14.4	10.7	5.9	3.5	(6)
(7)		DBStd	6.58	4.80	4.19	3.41	3.05	3.54	3.24	2.84	2.59	2.17	3.00	3.20	3.43	(7)
(8)		HDD10.0	1092	252	234	161	78	13	1	0	0	0	27	126	201	(8)
(9)		HDD18.3	3283	510	467	417	313	182	102	51	51	121	238	373	458	(9)
(10)		CDD10.0	913	0	0	2	14	93	161	233	228	131	48	3	0	(10)
(11)		CDD18.3	64	0	0	0	0	4	12	26	21	2	0	0	0	(11)
(12)		CDH23.3	402	0	0	0	1	39	77	158	117	10	0	0	0	(12)
(13)		CDH26.7	84	0	0	0	0	3	18	33	29	1	0	0	0	(13)
(14)	Wind	WSAvg	5.3	6.2	5.5	5.6	5.0	4.9	5.0	5.2	4.9	5.3	5.3	5.4	5.7	(14)
(15)	Precipitation	PrecAvg	841	66	55	51	40	58	83	93	86	83	73	75	77	(15)
(16)		PrecMax	1124	135	120	107	88	130	189	168	194	244	186	148	198	(16)
(17)		PrecMin	545	3	23	11	7	15	19	37	32	20	25	7	24	(17)
(18)		PrecStd	129	35	25	28	19	25	41	41	38	50	37	36	41	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.0	13.0	16.2	19.9	26.8	29.2	29.9	30.0	24.7	20.8	13.8	11.5	(19)
(20)			MCWB	9.2	10.5	11.3	12.3	16.9	20.0	20.5	20.1	18.7	16.7	11.4	9.7	(20)
(21)		2%	DB	9.3	10.4	12.5	17.1	24.1	25.2	27.2	26.6	21.3	18.1	12.2	10.4	(21)
(22)			MCWB	8.1	8.3	9.4	11.5	15.8	17.4	19.6	18.9	16.8	14.7	10.5	9.2	(22)
(23)		5%	DB	8.3	8.3	10.7	15.0	21.6	22.9	24.9	23.7	19.3	16.2	11.2	9.4	(23)
(24)			MCWB	7.3	6.6	8.3	10.7	14.7	17.1	18.5	18.2	15.8	13.6	10.0	8.4	(24)
(25)		10%	DB	7.4	6.9	9.3	13.0	18.6	20.8	22.6	21.7	17.9	14.7	10.3	8.3	(25)
(26)			MCWB	6.4	5.4	7.5	9.8	13.7	16.2	17.9	17.4	15.1	12.6	9.2	7.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.7	10.7	11.5	13.9	18.0	21.1	21.6	21.0	19.3	17.3	12.2	10.2	(27)
(28)			MCDWB	10.7	12.8	15.3	17.7	24.7	27.1	26.7	27.4	23.1	20.3	13.1	10.9	(28)
(29)		2%	WB	8.4	8.5	9.9	12.1	16.2	19.1	20.4	19.7	17.6	15.2	11.0	9.2	(29)
(30)			MCDWB	9.0	9.9	11.8	15.6	22.2	22.5	25.2	24.1	20.2	17.4	11.9	10.2	(30)
(31)		5%	WB	7.5	6.9	8.7	11.1	15.1	17.8	19.4	18.9	16.6	14.0	10.1	8.4	(31)
(32)			MCDWB	8.3	8.2	10.2	14.4	20.5	21.6	23.8	22.7	19.0	15.8	11.1	9.3	(32)
(33)		10%	WB	6.4	5.6	7.7	10.0	14.0	16.7	18.3	18.0	15.5	13.0	9.3	7.5	(33)
(34)			MCDWB	7.4	6.7	9.1	12.9	18.4	20.3	22.2	21.5	17.4	14.3	10.2	8.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	3.8	4.4	5.3	6.2	6.9	6.1	6.3	6.5	5.7	5.2	4.1	3.6	(35)
(36)			MCDBR	4.1	5.9	7.9	10.0	12.1	10.5	11.0	10.7	8.4	7.5	5.2	4.5	(36)
(37)			MCWBR	3.7	4.6	5.3	6.1	6.1	5.4	5.5	5.5	5.3	5.0	4.4	4.2	(37)
(38)		5% WB	MCDBR	4.0	5.7	7.0	8.8	10.8	9.1	9.5	8.9	7.7	6.4	5.1	4.3	(38)
(39)			MCWBR	3.8	4.8	5.1	5.7	5.9	5.3	5.4	5.1	5.5	4.8	4.5	4.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.327	0.346	0.376	0.410	0.405	0.404	0.420	0.413	0.388	0.362	0.340	0.318	(40)	
(41)		taud	2.311	2.318	2.253	2.215	2.271	2.299	2.268	2.297	2.354	2.388	2.343	2.303	(41)	
(42)		Ebn at Noon	629	743	796	817	843	847	824	806	779	716	607	560	(42)	
(43)		Edh at Noon	61	81	106	125	125	123	125	115	96	76	59	51	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.57	1.24	2.43	4.02	5.02	5.25	5.01	4.15	3.00	1.67	0.71	0.43	(44)	
(45)		RadStd	0.06	0.17	0.25	0.48	0.55	0.42	0.67	0.30	0.36	0.18	0.07	0.03	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (47 neighbors)		+0.33	N/A	N/A	N/A	N/A	N/A	N/A	-133	N/A	N/A

Nomenclature: See separate page